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CLASS	BSCS
SEMESTER	1st
SECTION	(A)
SUBJECT	C++
SUBMITTED TO	SIR ASAD
ASSIGNMENT LAB	4

Task - 1

⇒ Write a program to print positive number entered by the user. If the user enters negative number print number number entered is positive otherwise print number is negative.

```
#include <iostream>
  using namespace std;
  int main() {
    int num;
  cout << "Enter a number:";
  cin >> num;
  if(num>=0){
  cout<< "Number entered is positive" <<endl;
  } else {
     cout<< "Number is negative" <<endl;
  }
    return 0;
  }
```

Output:-

Enter a number :5
Number entered is positive
Enter a number:- 3
Number is negative

```cpp
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter a number:";
    cin>>num;
    if(num>=0){
        cout<<"Number entered is positive"<<endl;
    } else{
        cout<<" Number is negative"<<endl;
    }
    return 0;
}
```

```
Enter a number:5
Number entered is positive

[Program finished]
```

Task-2

→ program to check whether an integer is positive, negative or zero.

```
#include <iostream>
using namespace std;
int main() {
  int num;
cout << "Enter an integer: ";
cin >> num;
if (num>0) {
  cout << "Number is positive" << endl;
} else if (num<0) {
cout << "Number is negative" << endl;
}else {
cout << "Number is zero" << endl;
}
  return 0;
}
```

output :-

```
Enter an integer: 0
Number is zero
Enter an integer: 10
Number is positive
```

```cpp
#include <iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter an integer:";
    cin>>num;
    if(num>0){
        cout<<"Number is positive"<<endl;
    } else if (num<0){
        cout<<" Number is negative"<<endl;
    } else {
        cout<<"Number is zero"<<endl;
    }
    return 0;
}
```

```
Enter an integer:0
Number is zero

[Program finished]
```

Task - 3

⇛ Input : Mark
process : if mark greater than and equal to 75, score will be A. if mark less than 75 and greater than and equal to 60, score will be B. if mark less than 60 and greater than and equal to 45, score will be C. if mark less than 30, score will be D. Output print the grade of your score.

```
#include<iostream>
using namespace std;
int main() {
  int mark;
  cout<<"Enter your marks:";
  cin>>mark;
if(mark>=75){
cout<<"Grade A"<<endl;
}else if(mark>=60){
cout<<"Grade B"<<endl;
}else if(mark>=45){
cout<<"Grade C"<<endl;
} else if(mark>=30) {
cout<<"Grade D"<<endl;
 } else {
      cout<<"fail"<<endl;
   }
   return 0;
}
```

Output :-

```
Enter your marks:80
Grade A
Enter your mark:65
Grade B
Enter your mark:25
Grade D
```

```cpp
#include <iostream>
using namespace std;
int main(){
    int mark;
    cout<<"Enter your mark:";
    cin>>mark;
    if(mark>=75){
        cout<<"Grade A"<<endl;
    } else if (mark>=60){
        cout<<" Grade B"<<endl;
    } else if (mark>=45){
        cout<<"Grade C"<<endl;
    } else if (mark>=30){
        cout<<"Grade D"<<endl;
    } else {
        cout<<"Fail"<<endl;
    }
    return 0;
}
```

```
Enter your mark:80
Grade A

[Program finished]
```

Task_4

(2) "Find largest number & using nested if...else"

```
#include<iostream>
using namespace std;
int main(){
 int a,b,c;
 cout<<"Enter three numbers:";
 cin>>a>>b>>c;
 if(a>=b){
  if(a>=c)
   cout<<"Largest is:"<<a;
 else
  cout<<"Largest is:"<<c;
} else{
  if(b>=c)
   cout<<"Largest is:"<<b;
  else
   cout<<"Largest is:"<<c;
}
  return 0;
  }
```

output:-

```
Enter three numbers: 10 25 15
Largest Number is: 25
```

```cpp
#include <iostream>
using namespace std;
int main(){
    int a,b,c;
    cout<<"Enter three numbers:";
    cin>>a>>b>>c;
    if(a>=b){
        if(a>=c)
        cout<<"Largest is:"<<a;
    else
        cout<<" Largest is:"<<c;
    } else{
        if(b>=c)
        cout<<"Largest is :"<<b;
    else
        cout<<"Largest is :"<<c;
    |
    }
    return 0;
}
```

```
Enter three numbers:10 25 15
Largest is :25
[Program finished]
```

(2) "Check whether the number entered by the user is positive or not. if it is positive then calculate how many digits the number have."

```
#include <iostream>
using namespace std;
int main(){
    int num, count = 0;
    cout << "Enter a number: ";
   cin >> num;
   if (num>0) {
       int temp=num;
     while(temp!=0){
      temp /= 10;
      count++;
   } cout << "Number is positive" << endl;
    cout << "Digits: " << count << endl;
} else{
    cout << "Number is not positive" << endl;
    }
return 0;
}
```

Output:

```
Enter a number to check digits:
12345
Number is positive
Total digits: 5
```

```cpp
#include <iostream>
using namespace std;
int main(){
  int num,count=0;
  cout<<" Enter a number:";
  cin>>num;
  if(num>0){
  int temp=num;
  while(temp!=0){
  temp/=10;
  count ++;
}
  cout<<"Number is positive"<<endl;
  cout<<"Digits:"<< count<<endl;
} else{
  cout<<"Number is not positive"<<endl;
   }
    return 0;
}
```

```
 Enter a number:12345
Number is positive
Digits:5

[Program finished]
```